

# Player Feedback Analysis in Video Games

Evaluated Independent Data Analysis Project



Mohammad Kian Zohourian

April 2026

# Table of Contents

|                                                        |    |
|--------------------------------------------------------|----|
| 1. Introduction .....                                  | 3  |
| 2. Purpose of Study .....                              | 3  |
| 3. Data, Metadata, and Variables .....                 | 4  |
| 4. Methodology .....                                   | 7  |
| 4.1. Data Cleaning .....                               | 7  |
| 4.2. Feature Engineering and Selecting Variables ..... | 8  |
| 5. Analysis and Findings .....                         | 12 |
| 5.1. Descriptive Statistics .....                      | 12 |
| 5.2. Graphical Analysis .....                          | 14 |
| 5.3. Correlation Analysis .....                        | 23 |
| 5.4. Regression Analysis .....                         | 24 |
| 6. Discussion and Evaluation .....                     | 31 |
| 7. Conclusion .....                                    | 32 |
| 8. References .....                                    | 33 |

# 1. Introduction

Just as books, painting, theater, music, photography, and cinema have long held importance in literature, culture, storytelling, and artistic and literary movements, video games are no longer merely a means of entertainment or passing time. They now encompass new ideas, art forms, narratives, and sometimes even acts of rebellion. Due to the modern player's influence, their ability to make decisions, and receive immediate feedback, this medium—leveraging new technologies—has evolved into a new form of art.

Steam, developed by Valve, is the world's largest digital PC game store, distribution service, social network, and gaming platform. Players can purchase games, download and update them, message friends, broadcast gameplay, and participate in community workshops.

In this project, factors influencing positive user engagement with games on the Steam platforms such as leaving positive reviews and ultimately recommending games to other players were analyzed using the Steam API and Steam Spy API. However, it should be noted that this study is limited to the variables provided by the Steam API and does not include all factors that contribute to a game's popularity and reputation.

Since video games as a new artistic style and narrative method are a relatively recent phenomenon, they require deeper investigation. Nevertheless, this study presents a scientific analysis based on concrete and measurable factors.

## 2. Purpose of Study

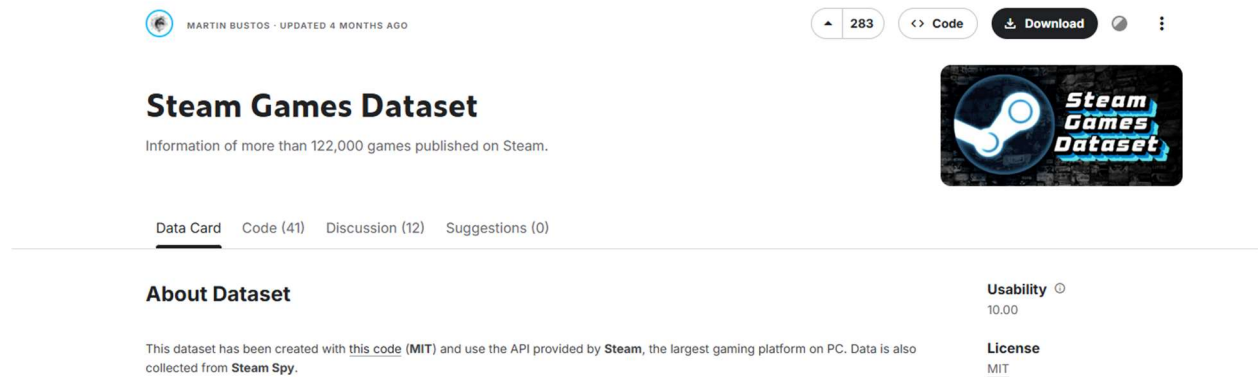
In the video game market, one of the most important factors that give a game recognition is its being recommended by media, platforms, and other players. The aim of the project is to determine, for a game on a platform (in this example, the Steam platform itself), as a result of which reasons and factors it:

1. Gains a good review and experience from players.
2. Is recommended to other players.

On the Steam platform, whether the majority of the reviews a game receives are positive or negative leaves a great impression on the player. In the research, data such as the number of reviews each game receives, positive and negative, as well as price and playtime, were used in order to find the factors that most lead to a good review, and ultimately to identify the elements that present a game as worth playing by other players.

# 1. Data, Metadata, and Variables

The dataset has been provided by Valve (developers of the Steam platform) through official APIs and the SteamSpy API based on these APIs, by MIT students to assist players and researchers, and can be downloaded via the Kaggle and Hugging Face platforms.



The dataset includes pricing information of games, user interaction (reviews, playtime), technical features, and genre information. The data was first created in 2022 and was last updated in January 2026.

Some games do not have a Metacritic score, and since the prices are the current discounted prices, the actual price needs to be determined manually. The data is entirely generated by APIs and there is likely no high-level control over the data itself but rather over the algorithms, and it is not supervised.

The data originates from real-world sources and exhibits characteristics such as high variance, skewed distribution, and zero-inflation. The dataset contains 122,611 observations and 40 variables, covering games from 1997 to 2026. Most variables in the data show right skewness and zero-inflation, and some variables are not exact but estimated data.

Variables included in the dataset:

| <i>Feature</i>                    | <i>Description</i>              | <i>Type</i>          |
|-----------------------------------|---------------------------------|----------------------|
| <i>AppID</i>                      | Unique identifier               | String (Numerical)   |
| <i>Name</i>                       | Game name                       | String (Categorical) |
| <i>Release date</i>               | Release date                    | String (Categorical) |
| <i>Estimated owners</i>           | Estimated range of owners       | String (Categorical) |
| <i>Peak CCU</i>                   | Peak concurrent players         | Int (Numerical)      |
| <i>Required Age</i>               | Minimum age required            | Int (Numerical)      |
| <i>Price</i>                      | Price in USD                    | Float (Numerical)    |
| <i>discount</i>                   | Discount rate (%)               | Int (Numerical)      |
| <i>DLC count</i>                  | Number of DLCs                  | Int (Numerical)      |
| <i>About the game</i>             | Information on the game         | String               |
| <i>Supported languages</i>        | Supported languages             | String (Categorical) |
| <i>Full audio languages</i>       | Full audio languages            | String (Categorical) |
| <i>Reviews</i>                    | Review data                     | Undefined            |
| <i>Header image</i>               | Header image URL                | String               |
| <i>Website</i>                    | Game website URL                | String               |
| <i>Support url</i>                | Support page URL                | String               |
| <i>Support email</i>              | Support email                   | String               |
| <i>Windows</i>                    | Windows support                 | Bool (Categorical)   |
| <i>Mac</i>                        | Mac support                     | Bool (Categorical)   |
| <i>Linux</i>                      | Linux support                   | Bool (Categorical)   |
| <i>Metacritic Score</i>           | Metacritic score                | Int (Numerical)      |
| <i>Metacritic url</i>             | Metacritic URL                  | String               |
| <i>Users score</i>                | User score                      | Int (Numerical)      |
| <i>Positive</i>                   | Number of positive votes        | Int (Numerical)      |
| <i>Negative</i>                   | Number of negative votes        | Int (Numerical)      |
| <i>Score rank</i>                 | Score ranking                   | Int (Numerical)      |
| <i>Achievements</i>               | Number of in-game achievements  | Int (Numerical)      |
| <i>Recommendations</i>            | User recommendations            | Int (Numerical)      |
| <i>Notes</i>                      | Additional information          | String               |
| <i>Average playtime forever</i>   | Average playtime (minutes)      | Int (Numerical)      |
| <i>Average playtime two weeks</i> | Average playtime (last 2 weeks) | Int (Numerical)      |
| <i>Median playtime forever</i>    | Median playtime                 | Int (Numerical)      |
| <i>Median playtime two weeks</i>  | Median playtime (last 2 weeks)  | Int (Numerical)      |
| <i>Developers</i>                 | Developer                       | String (Categorical) |
| <i>Publishers</i>                 | Publisher                       | String (Categorical) |
| <i>Categories</i>                 | Categories                      | String (Categorical) |
| <i>Genres</i>                     | Genres                          | String (Categorical) |
| <i>Tags</i>                       | Tags                            | String (Categorical) |
| <i>Screenshots</i>                | Screenshots URL                 | String               |
| <i>Movies</i>                     | Video URL                       | String               |

Despite the existence of this many observations and variables, most of these variables are unusable (since only numerical or categorical variables are utilized for this study), and because the data obtained from Steam is not very clean, further processing on the data is required. Additionally, measurement errors may exist in the variables. In this case:

|                                                                                                   |                                                             |
|---------------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| • <i>AppID:</i>                                                                                   | Only ID, not meaningful                                     |
| • <i>Release date:</i>                                                                            | Third-party estimate, may not be the actual release date    |
| • <i>Estimated owners:</i>                                                                        | Given as a range, not exact                                 |
| • <i>Peak CCU:</i>                                                                                | Not available for old/single-player games                   |
| • <i>Required age:</i>                                                                            | 0 = for everyone                                            |
| • <i>Price:</i>                                                                                   | Discounts affect the real price                             |
| • <i>DLC count:</i>                                                                               | Low importance                                              |
| • <i>Discount:</i>                                                                                | Seasonal, not fixed                                         |
| • <i>Supported languages, Full Audio Languages:</i>                                               | Text, categorical                                           |
| • <i>Reviews, Notes:</i>                                                                          | Text, for text mining analyses                              |
| • <i>Header image</i>                                                                             |                                                             |
| • <i>Website, Support url , Support email, Metacritic url, Screenshots, Movies, Header image:</i> | Link                                                        |
| • <i>Windows:</i>                                                                                 | Most games support                                          |
| • <i>Mac:</i>                                                                                     | Approximately 17% support                                   |
| • <i>Linux:</i>                                                                                   | Approximately 14%, not reliable (additional packages, mods) |
| • <i>Metacritic score, User score, score rank:</i>                                                | Missing in many games                                       |
| • <i>Positive, Negative, Recommendations: achievements:</i>                                       | Represents only a small portion of users                    |
| • <i>Average playtime forever, Median playtime forever:</i>                                       | Does not fully reflect playtime                             |
| • <i>Average playtime two weeks, Median playtime two weeks:</i>                                   | Very short-term                                             |
| • <i>Developers, Publishers:</i>                                                                  | Categorical but many different values                       |
| • <i>Categories, Genres, Tags:</i>                                                                | May have missing tags                                       |

## 2. Methodology

This research was conducted in the Python language using the Pandas, NumPy, statistics, SciPy, Matplotlib, Seaborn, Statsmodels, and Scikit-learn libraries. Since the data is real-world data, it has highly right-skewed variables, and not all variables are necessary for this research and can sometimes even be misleading. Therefore, outliers will first be removed to clean the data, and then functions will be applied to the variables in order to be able to use them.

### 4.1. Data Cleaning

Although the data may appear clean, there are variables and problems that needed to be cleaned. In some cases, data points that could create imbalance for this research or negatively affect the model that will be built and the analysis will be conducted were removed from the dataset. Such cases are as follows:

#### 1. **Pricing:**

Until the year 2026, the standard game price accepted by players and AAA game companies is \$70. That is, until 2026, the minimum price for the standard version of games (sold on Steam and forming the data points) is \$70.

In the dataset, there are games exceeding even \$1000; however, such values are few in number: (games more expensive than \$100  $\approx$  300 observations  $\approx$  0.25% of the data).

Games exceeding \$100 were removed from the dataset.

#### 2. **Games with no Owners:**

This study does not aim to examine all games available for sale on the platform, but rather the games that are played by players, in order to find the level of reactions or lack of reactions from players and the factors affecting them. For this reason, games that were never sold were removed.

As in the previous item, a relatively large number of observations were removed from the dataset (approximately 19,000 observations).

However, it should be noted that since Steam is an open platform, almost anyone can upload their game to this platform; therefore, many independent games exist on the platform without ever being purchased by anyone.

### 3. Multiple entries of the same game:

Some games were recorded again by the API at different times for different reasons (such as being assigned a new *AppID*). For example:

- On 1/11/2025, in the data retrieved by the API, game A with code 001 is recorded with x number of players and y number of recommendations.
- On 1/12/2025, in a new scan, instead of updating the previous information, the same game A is recorded as a separate observation with a new code (for example 039), with an updated number of players (x') and recommendations (y') due to an issue.

Therefore, by using the most up-to-date version of duplicated games, the extra recorded observations were removed from the dataset.

As a result of the cleaning, out of 122,611 observations, 99,927 observations were usable.

## 2.2. Feature Engineering and Selecting Variables

Variables used in the study are:

1. *Positive (dependent variable)*
2. *Median playtime forever*
3. *Achievements*
4. *Price*
5. *Release date*
6. *Genre*

Other variables were not used for various reasons; URLs such as movies, screenshots, website or texts such as *reviews* and *notes* cannot be used in this type of study (since they are large text and require text mining). *Estimated owners*, being from a 3rd-party API, may sometimes be incorrect and only represents a range, not an specific value (from '0-2000' to '100,000,000 – 200,000,000'). Both *Average playtime* variables show deviation in very short (such as narrative games) and very large games (online games, MMO). *Negative* and *Recommendations* naturally, increase when positive increases since these three variables are all in one category (votes and reviews). The same logic could be applied to the number of owners and players of a game.



|                 | Positive | Negative | Recommendations | Owneravg |
|-----------------|----------|----------|-----------------|----------|
| Positive        | 1.00     | 0.82     | 0.90            | 0.74     |
| Negative        | 0.82     | 1.00     | 0.74            | 0.77     |
| Recommendations | 0.90     | 0.74     | 1.00            | 0.63     |
| Owneravg        | 0.74     | 0.77     | 0.63            | 1.00     |

*Positive, Negative, Recommendation and Owneravg's  
(Estimated owners' median) correlation matrix*

Other variables were not used due to the errors and inconsistencies indicated in the data description section. In order to better utilize the variables, instead of using them as they are, functions are applied to make them usable.

Applied transformations:

**1. Real price:**

In the dataset, the Price variable does not represent the real price, but discounted prices. Using the Discount variable, the real prices is calculated and so emerges the *Real\_price* variable.

$$Real\_price = Price * (1 + (Discount)/100)$$

**2. From date to year:**

The year was extracted from the *Release date* variable to create the *Release\_year* variable. The year value was also calculated as variables from 0 to 29 (0 = 1997, 29 = 2026), though the results of the regression studies did not change, so *Release\_year* remained in the format of the year of release (1997, 2026).

**3. log1p for skewed data:**

Since the skewness of *Achievements*, *Median playtime forever*, and *Positive* variables are very high, the log1p of their values have been used. Because there is a zero-review problem in these values, the data is right-skewed. Since the aim of the study is to examine factors affecting positive reviews and it is not an exact count, this transformation is applied to the dependent variable.

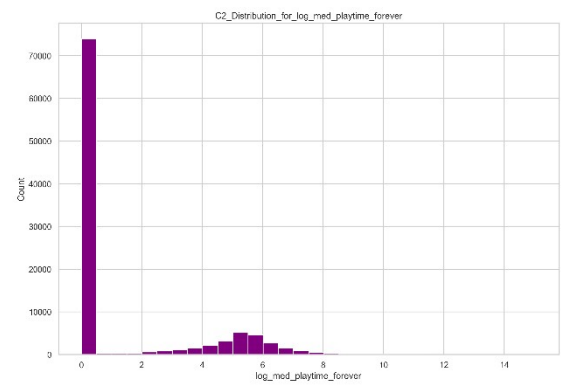
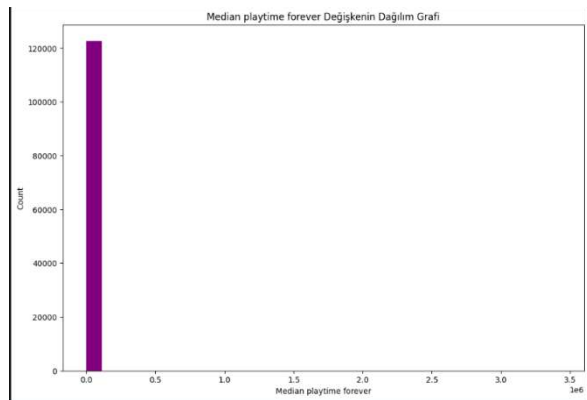
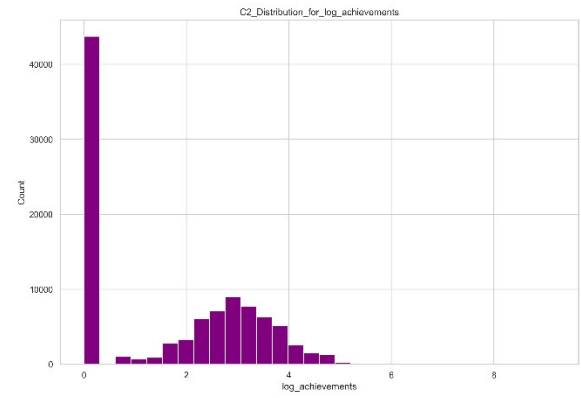
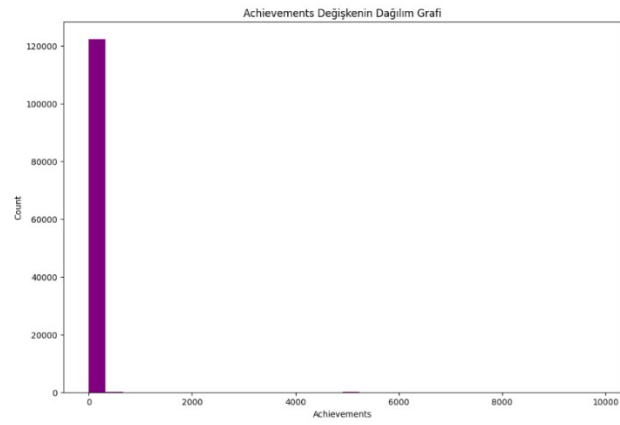
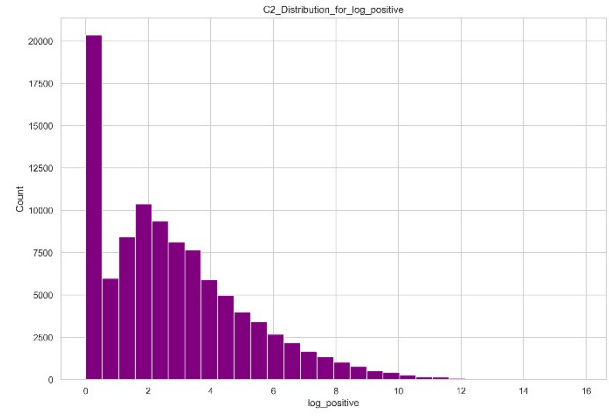
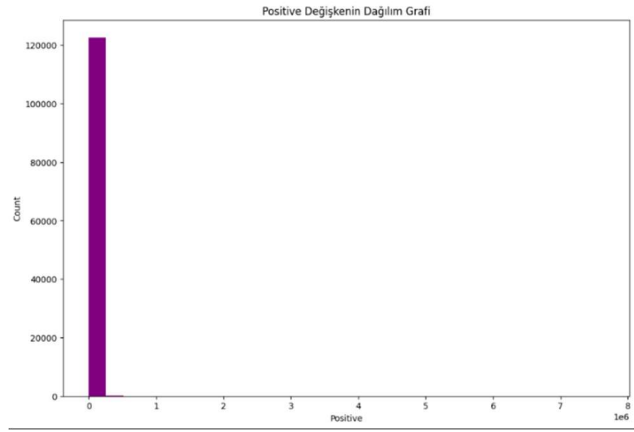


Figure 1: Distribution graphs of *Positive*, *Median playtime forever*, and *Achievements* and their log1p distributions

#### 4. Creating One-Hot Encoding variables from game genres:

In the dataset, *genre* is prepared as a list of genres associated with a game. In order to utilize this categorical variable and examine whether a game genre has a relationship with being positively perceived and recommended, columns were created one-by-one from the 10 most repeated game genres in the dataset (*Indie, Casual, Action, Adventure, Simulation, Strategy, RPG, Early Access, Sports, Racing*), assigning 1 if the game belongs to that genre, otherwise 0.

Example:

| <i>genre</i>    |     | <i>Racing</i> | <i>Action</i> | <i>Indie</i> |
|-----------------|-----|---------------|---------------|--------------|
| [Indie, Action] | = > | 0             | 1             | 1            |
| [Racing]        |     | 1             | 0             | 0            |
| [Action]        |     | 0             | 1             | 0            |

In order to prevent the resulting table from becoming too large, only the 10 most popular genres were selected. These 10 genres appear in 94.05% of the data. (Total genre word usage = 329,318, usage of selected 10 genres = 309,739)

| Genres                | count  | Genres                | count | Genres         | count |
|-----------------------|--------|-----------------------|-------|----------------|-------|
| Indie                 | 80,630 | Utilities             | 1,077 | Sexual Content | 88    |
| Casual                | 50,210 | Design & Illustration | 580   | Accounting     | 30    |
| Action                | 46,220 | Animation & Modeling  | 477   | Movie          | 2     |
| Adventure             | 45,141 | Education             | 461   | Documentary    | 1     |
| Simulation            | 24,114 | Violent               | 414   | Episodic       | 1     |
| Strategy              | 22,400 | Video Production      | 339   | Short          | 1     |
| RPG                   | 20,972 | GameDevelopment       | 320   | Tutorial       | 1     |
| Free ToPlay           | 12,172 | Gore                  | 257   | 360 Video      | 1     |
| EarlyAccess           | 11,091 | Audio Production      | 225   |                |       |
| Sports                | 4,882  | Software Training     | 219   |                |       |
| Racing                | 4,079  | Photo Editing         | 180   |                |       |
| Massively Multiplayer | 2,527  | Web Publishing        | 112   |                |       |

Count of each Genre keyword repetition

Final list of variables:

$y = \log\_positive$

$X = \log\_median\_playtime\_forever, Release\_year, \log\_achievements, Real\_price, Indie, Casual, Action, Adventure, Simulation, Strategy, RPG, Early Access, Sports, Racing$

(The *name* column is used as index)

## 5. Analysis and Findings

### 5.1. Descriptive Statistics

A series of calculations were performed for the selected variables: mean, median, mode, standard deviation, variance, coefficient of variation, skewness, and kurtosis.

|        | log_med_ | Release_y | log_achie | Real_price | Indie | Casual | Action | Adventure | Simulation | Strategy | RPG  | Early Acce | Sports | Racing | log_positiv |
|--------|----------|-----------|-----------|------------|-------|--------|--------|-----------|------------|----------|------|------------|--------|--------|-------------|
| mean   | 1.34     | 2021.03   | 1.67      | 5.73       | 0.71  | 0.43   | 0.41   | 0.40      | 0.21       | 0.20     | 0.18 | 0.10       | 0.04   | 0.04   | 2.76        |
| median | 0.00     | 2022.00   | 1.95      | 3.74       | 1.00  | 0.00   | 0.00   | 0.00      | 0.00       | 0.00     | 0.00 | 0.00       | 0.00   | 0.00   | 2.40        |
| std    | 2.37     | 3.33      | 1.62      | 7.07       | 0.45  | 0.50   | 0.49   | 0.49      | 0.41       | 0.40     | 0.39 | 0.30       | 0.20   | 0.19   | 2.38        |
| var    | 5.64     | 11.09     | 2.62      | 49.93      | 0.21  | 0.25   | 0.24   | 0.24      | 0.17       | 0.16     | 0.15 | 0.09       | 0.04   | 0.03   | 5.64        |
| CV     | 1.78     | 0.00      | 0.97      | 1.23       | 0.64  | 1.15   | 1.20   | 1.23      | 1.94       | 2.03     | 2.11 | 3.06       | 4.75   | 5.21   | 0.86        |
| skew   | 1.40     | -0.92     | 0.26      | 3.62       | -0.93 | 0.27   | 0.37   | 0.41      | 1.42       | 1.53     | 1.63 | 2.73       | 4.54   | 5.02   | 0.92        |
| kurt   | 0.36     | 0.90      | -1.17     | 23.40      | -1.14 | -1.92  | -1.86  | -1.83     | 0.02       | 0.35     | 0.67 | 5.47       | 18.62  | 23.22  | 0.67        |
| mode   | 0.00     | 2024.00   | 0.00      | 0.00       | 1.00  | 0.00   | 0.00   | 0.00      | 0.00       | 0.00     | 0.00 | 0.00       | 0.00   | 0.00   | 0.00        |

(Descriptive statistics table for variables: mean, median, std, var, CV, skew, kurt, mode)

Each variable is studied individually:

- *log\_median\_playtime*

In *log\_median\_playtime*, the mean (1.34) is quite higher than the median (0.00). High skewness (1.40) and zero concentration (mode = 0) indicate that most purchased games have very low play time. Low kurtosis (0.36) means that playtime is spread over a wide range around the mean and the center is sparse (platykurtic).

High coefficient of variation (1.78) and standard deviation (2.37) show high relative variability in playtime across games.

- *Release\_year*

Most of the games recorded in the dataset are released in 2024 (mode = 2024). Low coefficient of variation (0.00) indicates that games are concentrated within a relatively narrow range in terms of year.

The data is slightly left-skewed and concentrated in more recent years (skew = -0.92).

- *log\_achievement*

The closeness of mean (1.67) and median (1.95) and low skewness (0.26) indicate that the data becomes relatively symmetric after the log1p transformation. Kurtosis (-1.17) shows that the data is spread widely around the mean and has few extreme values. However, there is still zero concentration (mode = 0).

- *Real\_price*

Zero concentration (mode = 0) and high right skewness (skew = 3.6, mean > median > mode) show that most games are free or cheap. Despite trimming prices during data cleaning, extreme values still exist with heavy tails (kurt = 23.40). At the same time, there is high variability in prices (CV = 1.23, var = 49.92).

- Genres (*Indie, Casual, vs.*)

Since these variables are categorical, the most useful statistic is the mode. Among the selected 10 most frequent genres, the most common is Indie (mode = 1, number of Indie games = 80,630). Comparing means, after Indie, the most common genres are Casual, Action, and Adventure.

- *log\_positive*

The dependent variable is highly right-skewed (mean = 2.75 > median = 2.39 > mode = 0, skew = 0.91), relatively flat, with values spread around the mean and few extreme values. This indicates that most games receive no positive reviews, and among those that do, only a small number receive very high positive reviews, and those rare games pull the average upward.

Interaction among games is highly heterogeneous (std = 2.37, var = 5.64, CV = 0.91)

- General interpretation

There are high right skewness and zero-inflation in the game dataset. A small number of very popular games pull the averages upward. This structure creates difficulty for classical linear models and makes log transformation necessary.

## 5.2. Graphical Analysis

This section is dedicated to the study of the plotted graphs from the selected variables. Three types of graphs have been chosen: Distribution, Scatter, and Box Graphs.

- Distribution Graphs

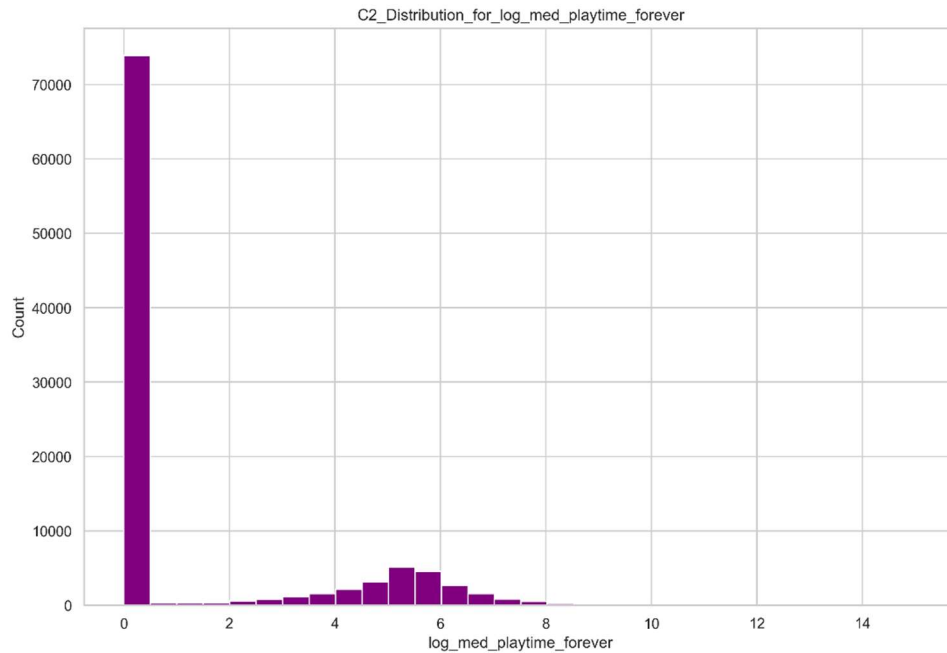


Figure 2: *log\_median\_playtime\_forever* distribution graph

The largest zero-density (among the selected variables) belongs to this variable. The median playtime of most games is either zero or very small. That is, when games are purchased, they are mostly either barely played or not played at all.

The observations in the middle of the graph have a long tail and high variability. Playtime varies greatly from game to game depending on players.

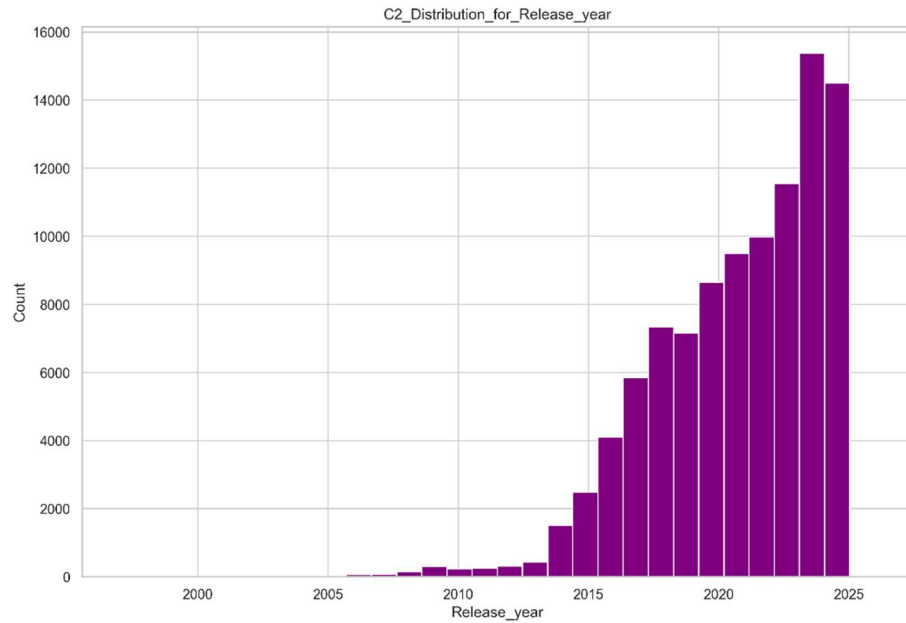


Figure 3: *log\_median\_playtime\_forever* distribution graph

Over time, a very interesting increase in the number of games can be observed. The first major change appears in 2014. Changes in Valve’s publishing policy between 2012–2013 and the availability of free or cheap game development tools caused more independent games to enter the platform.

The most common release year in the dataset is 2024.

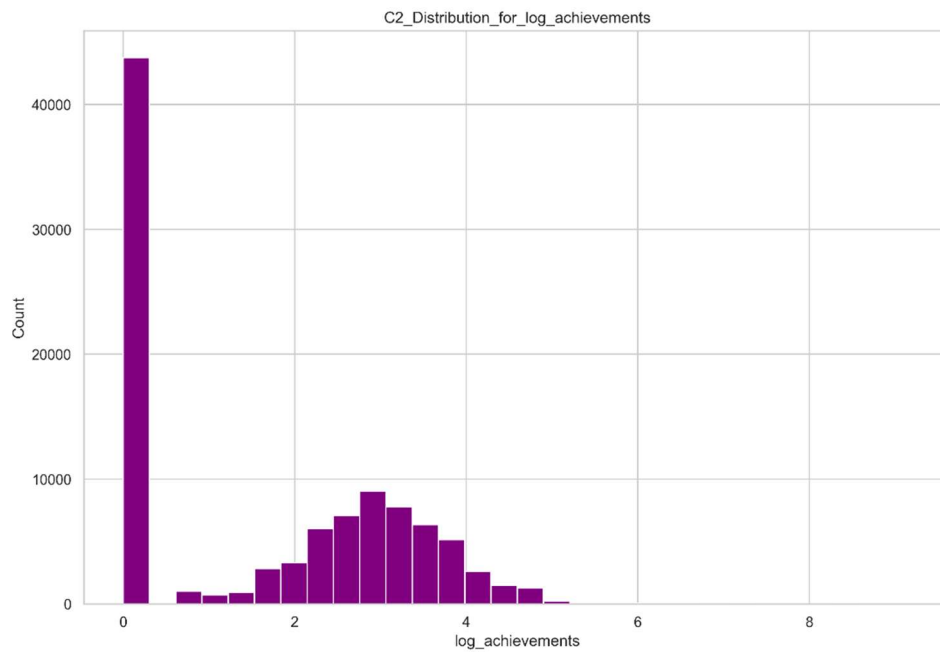


Figure 4: *log\_achievements* distribution graph

In this variable, as identified earlier, there are many zero values. Most games do not contain in-game achievements. Games with achievements show a shape resembling a bell curve. It can be said to be relatively normally distributed.

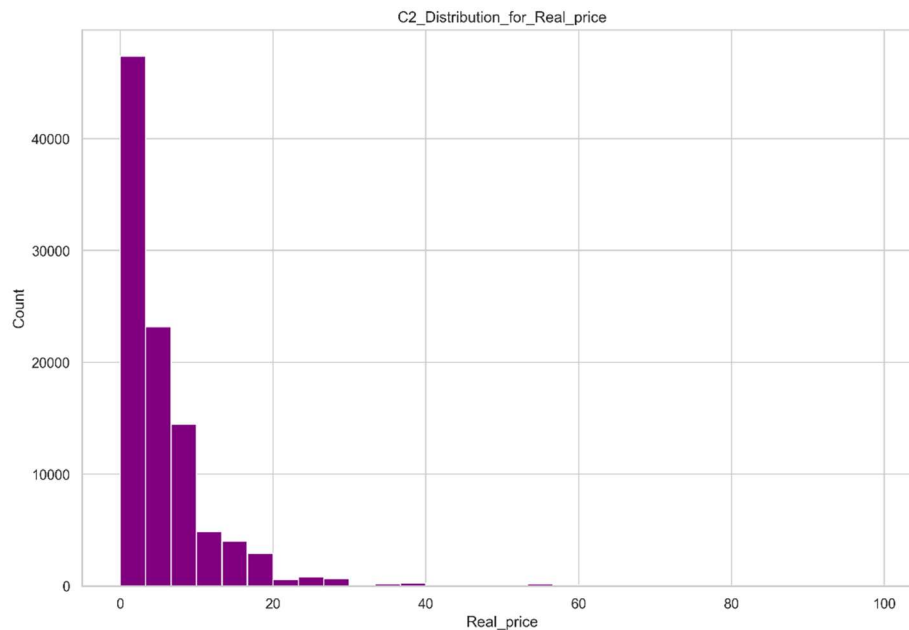


Figure 5: *Real\_price* distribution graph

The strong right skew shows that most games are free or very cheap. Very high-priced games exist but are few. Although prices in the gaming world are often in the \$20–\$60 range, games in that range are almost invisible in the population. Outliers exist but are few.

This graph shows how the price barrier in gaming has been reduced in this period and may indicate that video games have become more accessible to players (if they have the required equipment). At the same time, the large number of low-priced games reflects the presence of many independent games on Steam, since indie games are usually priced lower than AAA games.



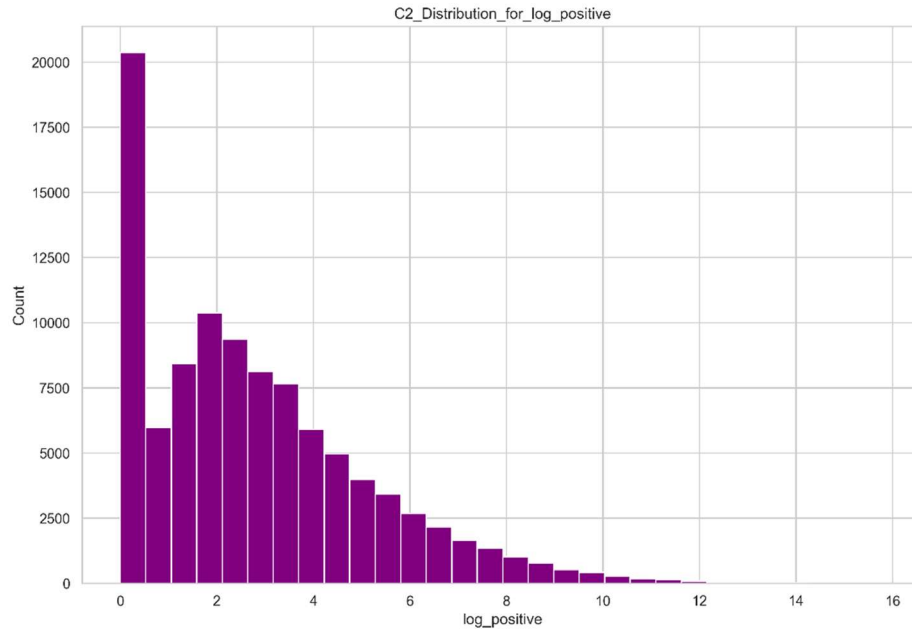


Figure 6: *log\_positive* distribution graph

Zero-density is still visible in this dependent variable. It is right-skewed with a long tail toward high values. About 20% of purchased games have received no positive reviews. This aligns with reality, since only a small portion of players leave reviews. Thus, many games remain without reviews. Most games receive very few positive reviews, but a very small number of massive games receive extremely high numbers of reviews, pulling the average upward. A small number of strong titles capture most of the market and revenue.

- Scatter Graphs

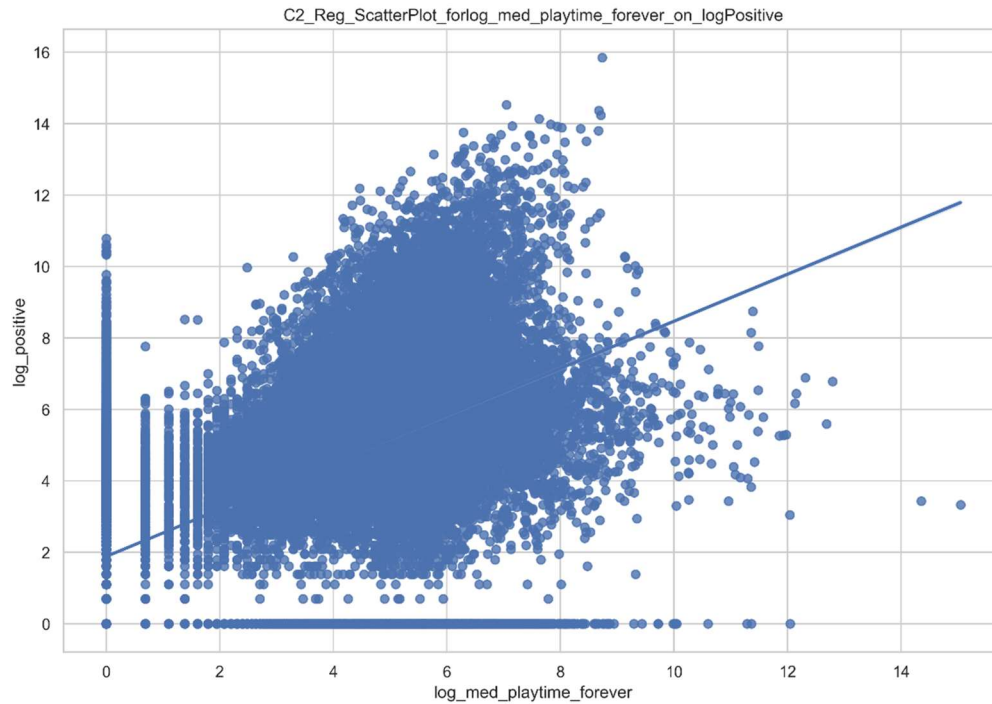


Figure 7: *log\_median\_playtime\_forever* scatter graph

Compared to other scatter plots, this shows the strongest relationship with positive reviews. As median playtime increases, the likelihood of receiving positive reviews also increases. However, the relationship is not perfectly linear and outliers exist.

Playtime strongly affects positive reviews up to a certain level (around 8–9 on the x-axis); beyond that, the effect weakens.

It cannot be definitively said that games keeping players engaged for long periods will receive very high positive reviews, but there is a strong relationship.

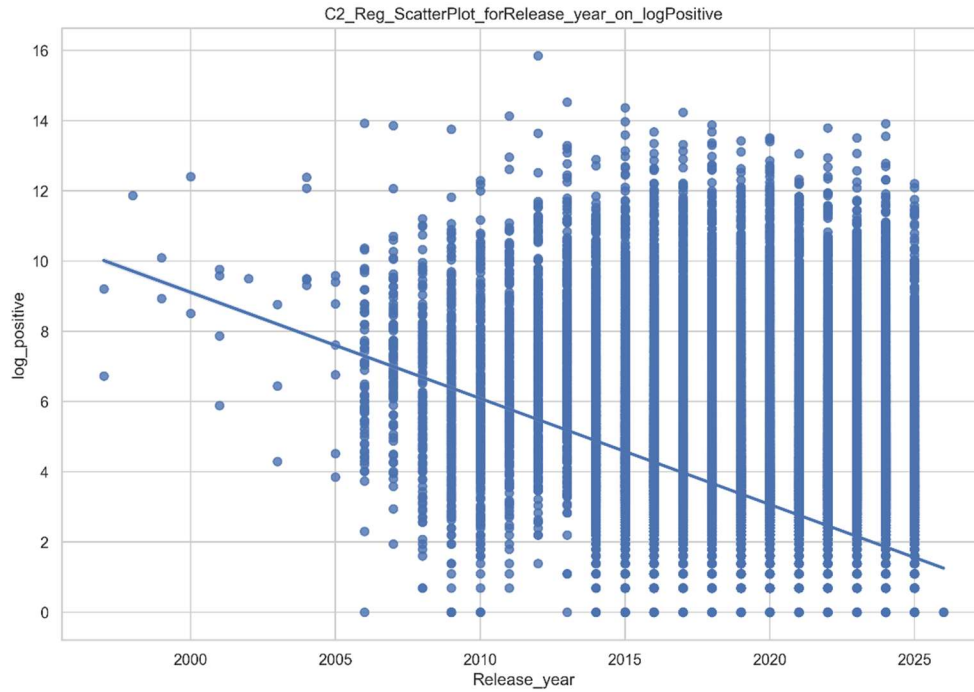


Figure 8: *Release\_year* scatter graph

A clear negative trend is visible. After the 2010s (especially the change seen in 2014), it has become harder for games to achieve high positive engagement over time. Older games had years to accumulate engagement and reviews. Newer games (especially indie games) appear to struggle more to stand out due to market saturation and recency.

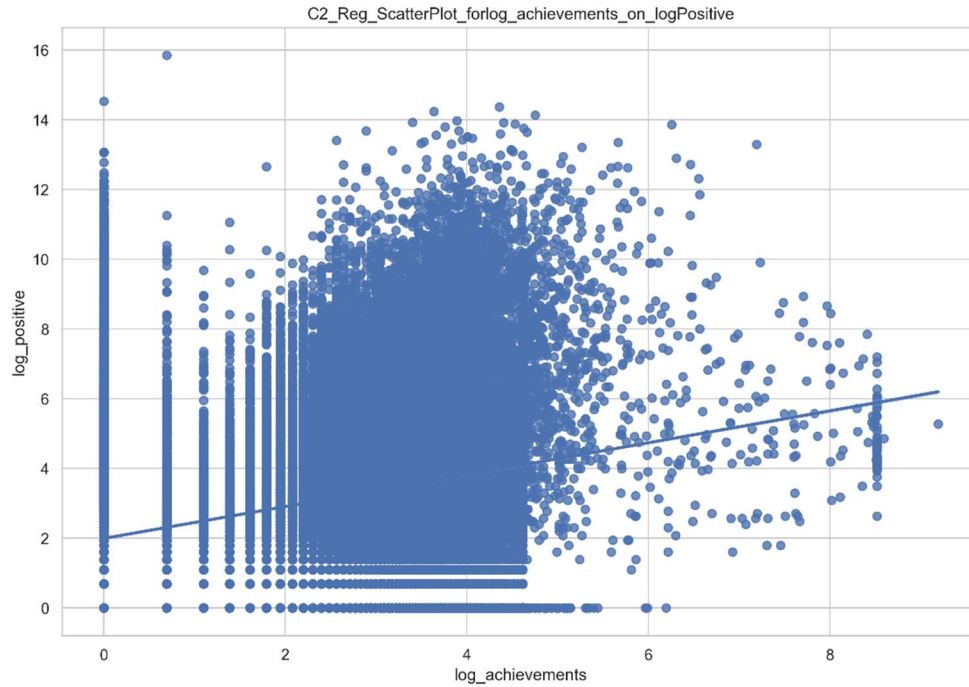


Figure 9: *log\_achievements* scatter graph

There is a positive correlation between achievements and positive reviews, but the data is more dispersed and appears in vertical clusters. Due to zero-density, many games without achievements still gain engagement. Therefore, high achievement count does not guarantee high engagement, though it has an effect.

Games with extremely high achievements show some positive reviews, but the games with the most positive reviews often have none or limited achievements. Despite relatively high variance, most games cluster within a certain range.

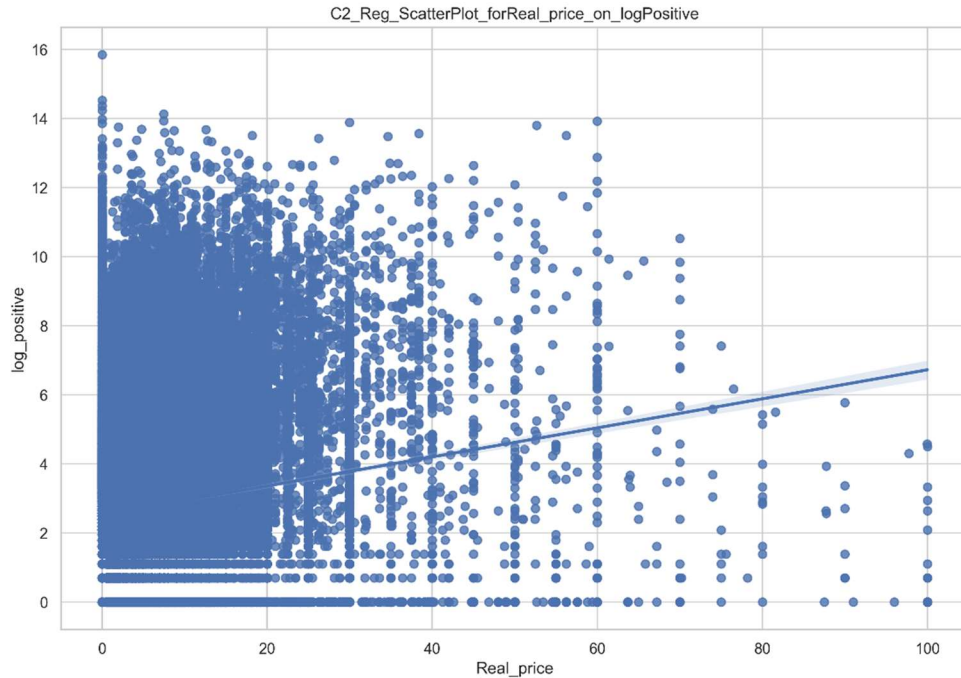


Figure 10: *Real\_price* scatter graph

There is not a strong relationship between price and engagement. A game's price does not directly determine whether it receives positive or negative evaluations, and user reviews on Steam do not appear to be strongly influenced by price.

After a certain price level, the number of games decreases. Games above \$60 do not reach the same engagement levels as more affordable games.

An interesting point is that the games with the highest number of positive reviews are free games. These have zero entry barriers and can reach very large audiences. Players who dislike them can leave without cost or review, while those who like them may be more inclined to leave positive feedback due to perceived "free quality content."

Because these games have large player bases, they may also receive many negative reviews, as previously noted (correlation between positive and negative = 0.82).

Most of these games are "Live Service" games, encouraging in-game spending.

- Box Graphs

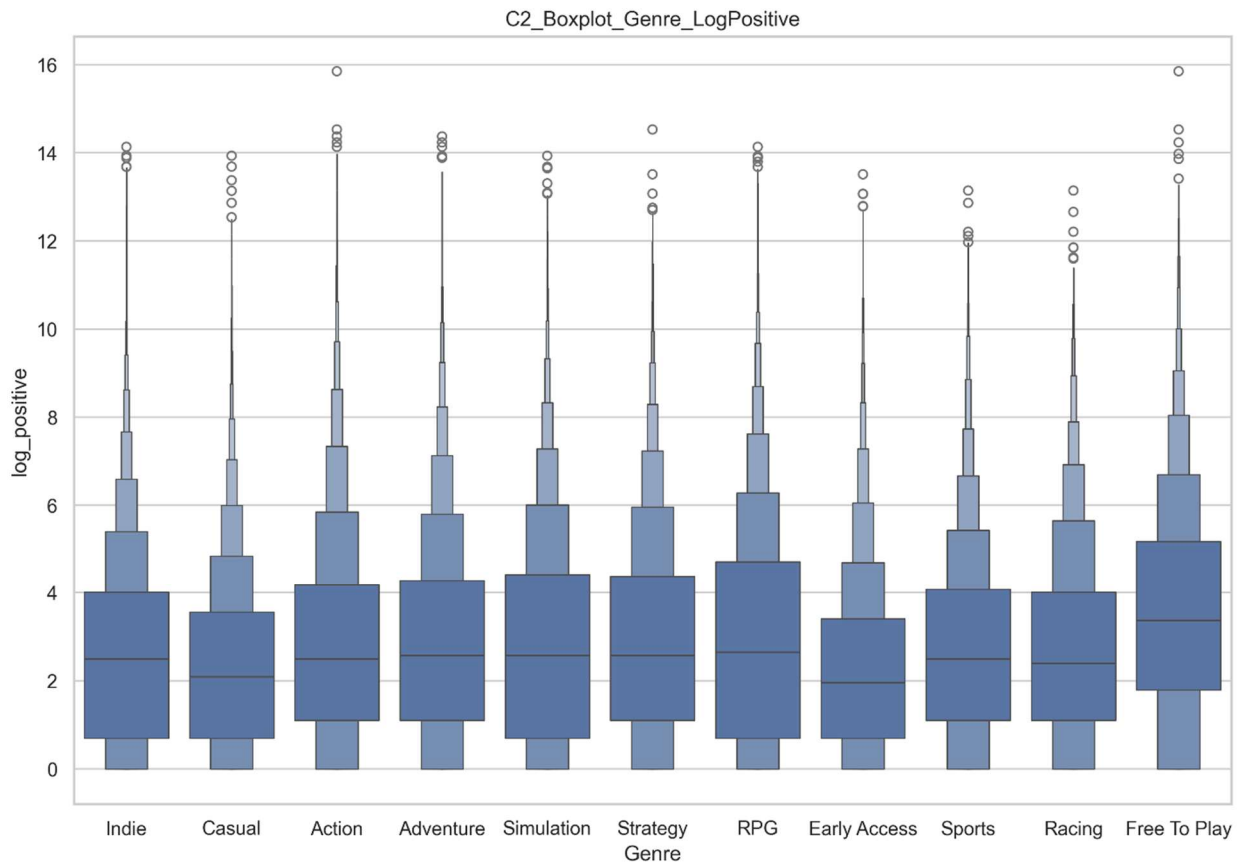


Figure 11: *Genre* box plot

To analyze game genres (including free games), genres were plotted against positive reviews. Based on medians, the highest positive review performance is likely in Free-to-Play, RPG, and Strategy games. Strategy, simulation, adventure, and action games appear to have nearly equal chances of receiving positive reviews.

Among the 10 most popular genres, Early Access and Casual games have the lowest number of positive reviews. Early Access games are usually incomplete and have fewer players. Casual games may have lower medians due to containing many low-quality (“shovelware”) titles.

Action and free games have the highest outliers. The most reviewed large-scale games on Steam are often from these categories. The Free-to-Play box is positioned higher than others, meaning it is harder for a free game to receive no reviews compared to a paid game.

### 5.3. Correlation Analysis

The Pearson correlation between all variables has been calculated

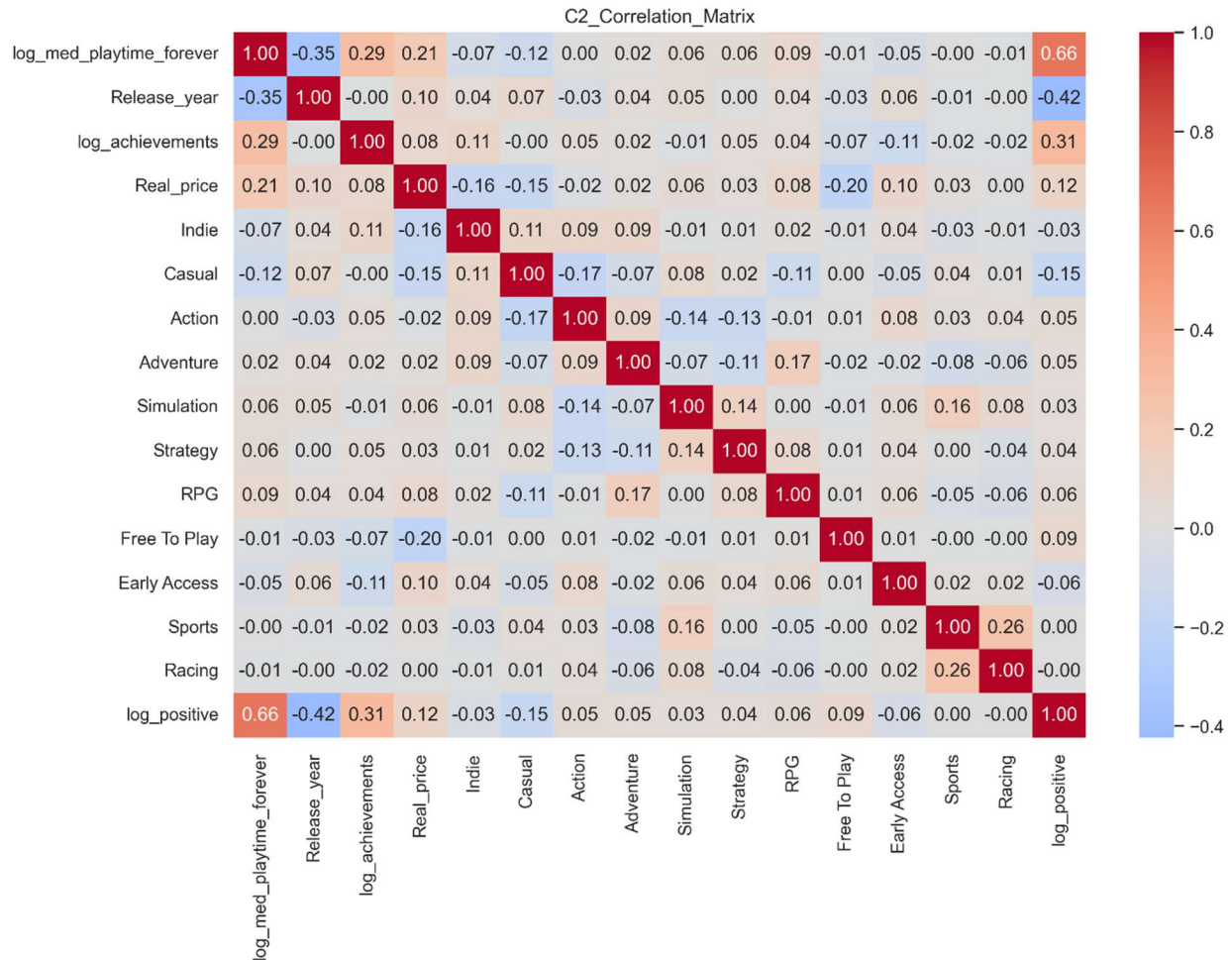


Figure 12: Correlation matrix heatmap

The variables with noticeable relationships with positive reviews are **playtime** (*log\_median\_playtime\_forever*: 0.66), **number of achievements** (*log\_achievements*: 0.31), and **release year** (*Release\_year*: -0.42).

The strongest and most influential signal is playtime. The small proportion of players who leave reviews are more likely to review games they play for longer periods. The number of achievements has a weak relationship. Many games do not have achievements at all, but in those that do, achievements may increase the likelihood of players continuing to play and leaving reviews.

The negative correlation with release year shows that newer games have fewer positive reviews. This may be due to the presence of many low-quality games among newer titles, or more likely

because newer games have not had enough time to accumulate reviews, unlike older games (accumulation bias).

Price has a very weak and positive relationship. Since most games on Steam are free or inexpensive, there may be a special situation for expensive games. AAA games being more expensive, having more followers and fans, and therefore receiving more reviews is a possibility.

Apart from the dependent variables, there are also interesting relationships among other factors. For example, a slight negative relationship between time and playtime (-0.35), the fact that games with more achievements are not necessarily played longer (only 0.29), and the relationship between Sports and Racing games (0.26).

## 5.4. Regression analysis

In order to mathematically express the factors affecting the number of positive reviews and the strength of the relationships between the selected variables, different regression models were constructed. Five different models were obtained.

In this study, four different regression models were tested. The first model examines the relationship between median playtime and positive reviews. The R-squared value of this model is approximately 43%. This result shows that playtime is a strong variable in explaining user engagement.

In the second model, the relationship between release year and positive reviews was examined. The explanatory power of this model was found to be approximately 18%. A negative relationship between release year and positive reviews was observed. This could be due to newer games not yet accumulating sufficient reviews.

In the third model, the relationship between achievements and positive reviews was analyzed. The R-squared value of this model is approximately 9.7%, indicating that while achievements have an effect on user engagement, they play a weaker role compared to playtime.

In the fourth model, the relationship between game price and positive reviews was examined. The explanatory power of this model is quite low, around 1.5%. This shows that price alone is insufficient to explain user engagement.

Finally, in the fifth model, a multiple regression model was constructed where all variables were considered together. In this model, playtime, release year, achievements, price, and game genres were evaluated together. The R-squared value of the model was found to be approximately 50%. This result shows that when variables are considered together, they explain user engagement better.

The outputs obtained from the models are as follows:



First model (*log\_med\_playtime\_forever* ve *log\_positive*):

| <i>Regression Statistics</i> |          |
|------------------------------|----------|
| Multiple R                   | 0.655874 |
| R Square                     | 0.43017  |
| Adjusted R Square            | 0.430164 |
| Standard Error               | 1.793462 |
| Observations                 | 99927    |

| ANOVA      |           |           |           |          |                       |
|------------|-----------|-----------|-----------|----------|-----------------------|
|            | <i>df</i> | <i>SS</i> | <i>MS</i> | <i>F</i> | <i>Significance F</i> |
| Regression | 1         | 242634.9  | 242634.9  | 75434.34 | 0                     |
| Residual   | 99925     | 321409.2  | 3.216504  |          |                       |
| Total      | 99926     | 564044.1  |           |          |                       |

|                          | <i>Coefficients</i> | <i>Standard Error</i> | <i>t Stat</i> | <i>P-value</i> | <i>Lower 95%</i> | <i>Upper 95%</i> | <i>Lower 95.0%</i> | <i>Upper 95.0%</i> |
|--------------------------|---------------------|-----------------------|---------------|----------------|------------------|------------------|--------------------|--------------------|
| Intercept                | 1.879622            | 0.006509              | 288.7518      | 0              | 1.866864         | 1.892381         | 1.866864           | 1.892381           |
| log_med_playtime_forever | 0.656234            | 0.002389              | 274.6531      | 0              | 0.651551         | 0.660918         | 0.651551           | 0.660918           |

The result of the R-squared test is 0.43; the median of playtime explains 43% of the variation in positive reviews. This is the highest value obtained among the simple models. The p-value is highly significant.

The obtained formula:

$$\log(\text{positive})=1.88+0.656\cdot\log(\text{med\_playtime})$$

Second model (*Release\_year* ve *log\_positive*):

| <i>Regression Statistics</i> |          |
|------------------------------|----------|
| Multiple R                   | 0.423453 |
| R Square                     | 0.179313 |
| Adjusted R Square            | 0.179304 |
| Standard Error               | 2.152327 |
| Observations                 | 99927    |

| ANOVA      |           |           |           |          |                       |
|------------|-----------|-----------|-----------|----------|-----------------------|
|            | <i>df</i> | <i>SS</i> | <i>MS</i> | <i>F</i> | <i>Significance F</i> |
| Regression | 1         | 101140.2  | 101140.2  | 21832.7  | 0                     |
| Residual   | 99925     | 462903.8  | 4.632513  |          |                       |
| Total      | 99926     | 564044.1  |           |          |                       |

|              | <i>Coefficients</i> | <i>Standard Error</i> | <i>t Stat</i> | <i>P-value</i> | <i>Lower 95%</i> | <i>Upper 95%</i> | <i>Lower 95.0%</i> | <i>Upper 95.0%</i> |
|--------------|---------------------|-----------------------|---------------|----------------|------------------|------------------|--------------------|--------------------|
| Intercept    | 613.454             | 4.133075              | 148.4256      | 0              | 605.3532         | 621.5548         | 605.3532           | 621.5548           |
| Release_year | -0.30217            | 0.002045              | -147.759      | 0              | -0.30618         | -0.29816         | -0.30618           | -0.29816           |

The result of the R-squared test is 0.17; the release year explains 17% of the variation in positive reviews. The p-value is significant.

The obtained formula:

$$\log(\text{positive}) = 613.45 - 0.302 \cdot \text{Release\_year}$$

Since the release year is defined as a very large numerical variable, the intercept value is very high.

Third model (*log\_achievements* ve *log\_positive*):

| <i>Regression Statistics</i> |          |
|------------------------------|----------|
| Multiple R                   | 0.311442 |
| R Square                     | 0.096996 |
| Adjusted R Square            | 0.096987 |
| Standard Error               | 2.25769  |
| Observations                 | 99927    |

| ANOVA      |           |           |           |          | <i>Significance</i> |
|------------|-----------|-----------|-----------|----------|---------------------|
|            | <i>df</i> | <i>SS</i> | <i>MS</i> | <i>F</i> | <i>F</i>            |
| Regression | 1         | 54710.06  | 54710.06  | 10733.43 | 0                   |
| Residual   | 99925     | 509334    | 5.097163  |          |                     |
| Total      | 99926     | 564044.1  |           |          |                     |

|                  | <i>Coefficients</i> | <i>Standard Error</i> | <i>t Stat</i> | <i>P-value</i> | <i>Lower 95%</i> | <i>Upper 95%</i> | <i>Lower 95.0%</i> | <i>Upper 95.0%</i> |
|------------------|---------------------|-----------------------|---------------|----------------|------------------|------------------|--------------------|--------------------|
| Intercept        | 1.992551            | 0.010263              | 194.1474      | 0              | 1.972435         | 2.012666         | 1.972435           | 2.012666           |
| log_achievements | 0.456733            | 0.004409              | 103.6023      | 0              | 0.448093         | 0.465374         | 0.448093           | 0.465374           |

The result of the R-squared test is 0.09; the median of playtime explains 9% of the variation in positive reviews. The p-value is significant, but the explanatory power is very low.

The obtained formula:

$$\log(\text{positive})=1.99+0.457\cdot\log(\text{achievements})$$

Fourth model (*Real\_price* ve *log\_positive*):

| <i>Regression Statistics</i> |          |
|------------------------------|----------|
| Multiple R                   | 0.12272  |
| R Square                     | 0.01506  |
| Adjusted R Square            | 0.01505  |
| Standard Error               | 2.357894 |
| Observations                 | 99927    |

| ANOVA      |           |           |           |          |                       |
|------------|-----------|-----------|-----------|----------|-----------------------|
|            | <i>df</i> | <i>SS</i> | <i>MS</i> | <i>F</i> | <i>Significance F</i> |
| Regression | 1         | 8494.629  | 8494.629  | 1527.903 | 0                     |
| Residual   | 99925     | 555549.4  | 5.559664  |          |                       |
| Total      | 99926     | 564044.1  |           |          |                       |

|            | <i>Coefficients</i> | <i>Standard Error</i> | <i>t Stat</i> | <i>P-value</i> | <i>Lower 95%</i> | <i>Upper 95%</i> | <i>Lower 95.0%</i> | <i>Upper 95.0%</i> |
|------------|---------------------|-----------------------|---------------|----------------|------------------|------------------|--------------------|--------------------|
| Intercept  | 2.519562            | 0.009606              | 262.3017      | 0              | 2.500735         | 2.538389         | 2.500735           | 2.538389           |
| Real_price | 0.041263            | 0.001056              | 39.0884       | 0              | 0.039194         | 0.043332         | 0.039194           | 0.043332           |

The result of the R-squared test is 0.01; the median of playtime explains 1% of the variation in positive reviews. This variable alone is insufficient to determine the number of positive reviews.

Fifth model (Multiple):

| <i>Regression Statistics</i> |             |
|------------------------------|-------------|
| Multiple R                   | 0.710167261 |
| R Square                     | 0.504337538 |
| Adjusted R Square            | 0.504268085 |
| Standard Error               | 1.672787244 |
| Observations                 | 99927       |

| ANOVA      |           |             |             |             |                       |
|------------|-----------|-------------|-------------|-------------|-----------------------|
|            | <i>df</i> | <i>SS</i>   | <i>MS</i>   | <i>F</i>    | <i>Significance F</i> |
| Regression | 14        | 284468.599  | 20319.18564 | 7261.475595 | 0                     |
| Residual   | 99912     | 279575.4732 | 2.798217163 |             |                       |
| Total      | 99926     | 564044.0722 |             |             |                       |

|                           | <i>Coefficients</i> | <i>Standard Error</i> | <i>t Stat</i> | <i>P-value</i> | <i>Lower 95%</i> | <i>Upper 95%</i> | <i>Lower 95.0%</i> | <i>Upper 95.0%</i> |
|---------------------------|---------------------|-----------------------|---------------|----------------|------------------|------------------|--------------------|--------------------|
| Intercept                 | 353.9089            | 3.5427                | 99.8978       | 0.0000         | 346.9653         | 360.8526         | 346.9653           | 360.8526           |
| log_med_play_time_forever | 0.5103              | 0.0026                | 195.162       | 0.0000         | 0.5052           | 0.5154           | 0.5052             | 0.5154             |
| Release_year              | -0.1743             | 0.0018                | -99.408       | 0.0000         | -0.1777          | -0.1709          | -0.1777            | -0.1709            |
| log_achievements          | 0.2325              | 0.0035                | 66.4235       | 0.0000         | 0.2256           | 0.2393           | 0.2256             | 0.2393             |
| Real_price                | 0.0054              | 0.0008                | 6.7523        | 0.0000         | 0.0039           | 0.0070           | 0.0039             | 0.0070             |
| Indie                     | 0.0029              | 0.0122                | 0.2391        | 0.8110         | -0.0209          | 0.0267           | -0.0209            | 0.0267             |
| Casual                    | -0.3116             | 0.0112                | -27.789       | 0.0000         | -0.3336          | -0.2896          | -0.3336            | -0.2896            |
| Action                    | 0.1127              | 0.0113                | 9.9920        | 0.0000         | 0.0906           | 0.1348           | 0.0906             | 0.1348             |
| Adventure                 | 0.1793              | 0.0112                | 16.0165       | 0.0000         | 0.1574           | 0.2013           | 0.1574             | 0.2013             |
| Simulation                | 0.1322              | 0.0135                | 9.7616        | 0.0000         | 0.1057           | 0.1588           | 0.1057             | 0.1588             |
| Strategy                  | 0.0407              | 0.0137                | 2.9649        | 0.0030         | 0.0138           | 0.0676           | 0.0138             | 0.0676             |
| RPG                       | 0.0278              | 0.0141                | 1.9687        | 0.0490         | 0.0001           | 0.0555           | 0.0001             | 0.0555             |
| Early Access              | -0.1115             | 0.0184                | -6.0638       | 0.0000         | -0.1476          | -0.0755          | -0.1476            | -0.0755            |
| Sports                    | 0.0011              | 0.0276                | 0.0393        | 0.9687         | -0.0530          | 0.0551           | -0.0530            | 0.0551             |
| Racing                    | 0.1131              | 0.0297                | 3.8065        | 0.0001         | 0.0549           | 0.1714           | 0.0549             | 0.1714             |

Based on the p-values, the p-values of Indie and Sports are above 0.05 and become insignificant. Apart from that, the other p-values are significant and can be used in the dataset.

Final Formula:

$$\begin{aligned}\log(\text{positive}) = & 353.91 \\ & + 0.51 \cdot \log(\text{med\_playtime}) \\ & - 0.17 \cdot \text{Release\_year} \\ & + 0.23 \cdot \log(\text{achievements}) \\ & + 0.005 \cdot \text{Real\_price} + \text{Additional effect of Genres}\end{aligned}$$

However, a noteworthy finding is that some variables that had high effects in simple models show reduced effects in the multiple model. For example, the coefficient of the strongest variable, playtime, decreased from 0.65 in the simple model to 0.51 in the multiple model. Similarly, the coefficient of the achievements variable also decreased significantly. This situation may indicate the presence of overlapping effects (confounding) between variables.

The same applies to the price variable; its effect is almost eliminated in the multiple model. This reveals that this variable contains an indirect relationship. This may be due to the fact that expensive games are generally more played and gain higher engagement.

When genre variables are examined, the negative effect of the Casual genre draws attention, while Action, Adventure, and Simulation genres are observed to have positive effects. This result shows that games with deeper and longer playability can create more engagement.

## 6. Discussion and Evaluation

In this study, the factors determining user engagement in Steam games were analyzed using statistical methods. On the Steam platform, it has been shown that playtime strongly affects the engagement and positive reviews a game receives. It was shown how game genres have an effect, the accumulation effect over time, and that engagement is not strongly affected by price.

An important finding of this study is that the dataset is not homogeneous. It was observed that a small number of games gather very high engagement, while the majority remain at low engagement levels. There are few successful games and many games that are not played. This situation indicates an imbalanced distribution structure in the digital game market, with right skewness and strong zero-inflation.

It is also shown that playtime alone is not sufficient to explain user engagement. In the multiple regression model, including other variables increased explanatory power, but this increase remained limited. This indicates that user engagement has a multidimensional structure. If variables such as the exact number of players were available in the dataset, they could be examined as an influential factor in the study.

The study has some limitations. Since the analysis is based on observational data, the relationships obtained cannot be interpreted as causal. Additionally, the skewness and zero-inflation in the dataset challenge classical regression assumptions. The exclusion of important variables such as marketing, user experience, and social interaction also limits explanatory power.

In this research, to measure the reactions and engagement a game receives, the  $\log_{1p}$  of the number of positive reviews was used. This may be a limitation of the study; while positive reviews are a good variable to understand how a game is perceived by players, they cannot cover everything. For example, Metacritic scores of the game, or its presentation and evaluation on social media and video game news platforms.

Moreover, in the gaming world, the number of players who take time to leave reviews is normally very low (approximately 5%). The recognition and strong engagement of a game are not determined only by the reviews it receives. Today, games are considered as literary and artistic works; therefore, it is beneficial to examine aspects such as the game's visual and artistic style, story, theme, lore, music, and its connections to other works (games adapted from books or films, or games connected to other games).

Finally, these positive reviews were obtained from only one platform, namely Steam. Although Steam is one of the largest platforms in the gaming world, it is still a single platform and targeted at a specific audience (those who play on PC). If data from different platforms (PS Store, Nintendo Store, Microsoft Store, Google Play) were used, a better and more general study for the gaming world could be conducted.

## 7. Conclusion

In this study, the factors determining user engagement in Steam games were analyzed using statistical methods. The findings show that playtime has the strongest effect on positive reviews. In addition, achievements and certain game genres were also observed to increase engagement. The model's approximately 50% explanatory power indicates that user engagement has a complex and multidimensional structure.

The negative effect of the release year variable is explained by the accumulation of engagement over time, while the effect of price was found to be limited. In the constructed models, playtime alone explains about 43% of the log of positive reviews and shows a strong correlation, demonstrating that it is a powerful factor with good standalone explanatory power. The more a game keeps a player engaged and the more time the player spends in it, the higher the likelihood of receiving feedback from the player. Online games (e.g., CS, Dota 2, etc.) are good examples of this situation.

Older games on the platform show a negative relationship between release year and positive reviews, partly due to pre-Steam Greenlight periods when higher-quality games were more likely to accumulate reviews over time. New games may be at a disadvantage due to time and the high number of competing titles in the market and therefore need to perform at a higher level.

When genres are examined, games that keep players engaged and make them part of the experience (RPG, Action, Simulation) and games with no entry price barrier (Free-to-Play) were shown to receive more reviews. Overall, user engagement is largely related to how long games are played and how much content they provide to the player. However, it is clear that this engagement cannot be fully explained by these variables alone, and more comprehensive analyses require additional variables.

To summarize, player engagement in video games is a multidimensional factor, in relation to multiple variables all at once, Some measurable and some not in the traditional sense of numerical records, in an oversaturated market with a few "hits".



## 8. References

- Steam - Valve Announces Steam Greenlight  
[<https://store.steampowered.com/oldnews/8367>]
- The Guardian - Valve kills Steam Greenlight – here's why it matters  
[<https://www.theguardian.com/technology/2017/feb/13/valve-kills-steam-greenlight-heres-why-it-matters#:~:text=The%20Valve%20Downed%20Steam%20store%20used%20a%20program,would%20be%20allowed%20space%20on%20the%20store.>]
- Langan - CUMULATIVE EFFECTS  
[[https://gaming.ny.gov/system/files/documents/2025/10/27\\_free\\_feis\\_cumulative\\_20251009\\_tc\\_togc.pdf#:~:text=Introduction,term%20impacts%2C%20and%20synergistic%20effects.](https://gaming.ny.gov/system/files/documents/2025/10/27_free_feis_cumulative_20251009_tc_togc.pdf#:~:text=Introduction,term%20impacts%2C%20and%20synergistic%20effects.)]
- Wikipedia – Confounding  
[<https://en.wikipedia.org/wiki/Confounding#:~:text=In%20causal%20inference%2C%20confounding%20is%20a%20form,a%20spurious%20association%20between%20exposure%20and%20outcome.>]
- IBM - What is multicollinearity? [<https://www.ibm.com/think/topics/multicollinearity>]
- An Empirical Study of Game Reviews on the Steam Platform  
[[https://www.researchgate.net/publication/324923032\\_An\\_Empirical\\_Study\\_of\\_Game\\_Reviews\\_on\\_the\\_Steam\\_Platform](https://www.researchgate.net/publication/324923032_An_Empirical_Study_of_Game_Reviews_on_the_Steam_Platform)]
- Epyllion – The State of Video Gaming in 2025  
[[https://static1.squarespace.com/static/5d8e9007bc3d0e18a4c49673/t/69090f8751925142fff8dcef/1762201479663/Gaming2025\\_vNov.2.2025+-+COMPRESS.pdf](https://static1.squarespace.com/static/5d8e9007bc3d0e18a4c49673/t/69090f8751925142fff8dcef/1762201479663/Gaming2025_vNov.2.2025+-+COMPRESS.pdf)]
- Reddit - [OC] Nominal and inflation adjusted video game prices in the US since 1985 (Not source information, only a supplementary reference)  
[[https://www.reddit.com/r/dataisbeautiful/comments/136hw1k/oc\\_nominal\\_and\\_inflation\\_adjusted\\_video\\_game/](https://www.reddit.com/r/dataisbeautiful/comments/136hw1k/oc_nominal_and_inflation_adjusted_video_game/)]